

Total no. of Pages: 2

Roll no.

Summer Semester
END-SEMESTER EXAMINATION, July 2023

Course Code- EEEEC03/ICICC03/ECECC05

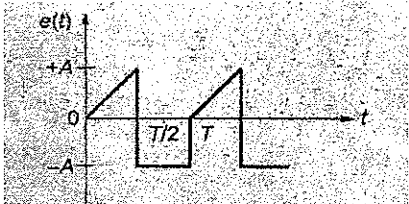
Course Title- Signal and System

Total No. of Students in Course: 28

Time- 3 Hours

Max. Marks- 50

Note: - Attempt all the five questions. Missing data/ Information if any, maybe suitably assumed & mentioned in the answer.

Q No	Question	Marks	CO
Q1	Attempt any 2 parts of the following		
1a	The energy of the continuous time signal is $x(t) = r(t-1)-r(t-2)-r(t-3)+r(t-4)$	5	1
1b	Determine the rms value of the periodic waveform $e(t)$ shown in the figure is 	5	1
1c	Define steps in Convolution. Determine the output $y(t)$ for a given continuous time LTI system with unit-impulse response $h(t)$, $x(t) = e^{-2t} \cdot u(t)$ $h(t) = u(t)$	5	1
Q2	Attempt any 2 parts of the following		
1a	Consider a continuous time system with input $x(t)$ and output $y(t)$ given by $y(t) = x(t) \cos(t)$ This system is linear or not and time-invariant or not.	5	1
1b	Find the Laplace transform of $x(t) = e^{2t} \sin 5t u(t)$	5	2
1c	Determine the partial fraction expansion and hence Inverse Laplace transform of the function, $X(s) = \frac{s^2 + 2s - 2}{s(s+2)(s+3)}, \text{ ROC: } \text{Re}(s) > 3$	5	2
Q3	Attempt any 2 parts of the following		
1a	Consider the LTI system given by the differential equation $\frac{dy}{dt} + \frac{1}{6}y(t) = 3x(t)$ The response of the system to the input $x(t) = 3e^{-\frac{t}{3}}u(t)$	5	2
1b	The laplace transform of the function $f(t)$ $F(s) = \frac{5s^2 + 23s + 6}{s(s^2 + 2s + 2)}$ As $t \rightarrow \infty$, $f(t)$ approaches	5	2

1c	Solve the differential equation using Inverse Laplace, $\frac{dy(t)}{dt} + 5y(t) = x(t)$ With initial condition $y(0+) = -2$, and input $x(t) = 3e^{-2t}u(t)$.	5	2
Q4	Attempt any 2 parts of the following		
1a	The z-transform of the signal $x(n)$, and its ROC, $x(n) = \begin{cases} a^n & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$	5	2
1b	Obtain the z-transform of the discrete time signal, and ROC $X[n] = u(-n)$	5	3
1c	Define sampling theorem. Determine the inverse z-transform of $F(z) = \frac{1}{z+1}$; if $u(t)$ is the unit step and $\delta(t)$ is the unit impulse function.	5	3
Q5	Attempt any 2 parts of the following		
1a	Determine the Fourier transform of the signal $x(t)$ $x(t) = \begin{cases} e^{-j10t} & \text{for } t \leq 1 \\ 0 & \text{for } t > 1 \end{cases}$	5	3
1b	A periodic function $f(t)$, with a period of 2π is represented as its fourier series, $F(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$ $f(t) = \begin{cases} A \sin t, & 0 \leq t \leq \pi \\ 0 & \pi \leq t \leq 2\pi \end{cases}$ The fourier coefficients a_1 and b_1 of $f(t)$ are:	5	4
1c	Obtain the fourier transform of the single side exponential function $e^{-at} \cdot u(t)$. Also draw the spectrum.	5	5